

I compared the baseline model and the fine-tuned model,

Baseline quantized model (original prepared test set): 92.5% correct (N=1113, 75 unknown).

Baseline model on my collected dataset (before fine-tuning): 56.0% accuracy (100 samples).

Fine-tuned model on my collected dataset: 95.0% accuracy (100 samples).

These updated results show that fine-tuning significantly improved performance on my own gesture style. The baseline model had a clear domain gap on personal data (56%), while fine-tuning reduced that gap and boosted accuracy to 95%, a +39 percentage-point improvement.

The improvement is likely because the model adapted to my specific motion patterns, stroke timing, and orientation variations captured in my collected samples. In contrast, the baseline model was trained on a different distribution of gestures and therefore generalized less effectively to my personal input style.

Conclusion: with the updated collected dataset, fine-tuning is clearly beneficial and should be used for deployment. The fine-tuned model is the better choice for my Lab 7 Arduino inference tests.

Codex is used for code debugging, writing logic checking and content summarizing.